A loop goes somewhere.

01 Why does the middle rise into a loop?



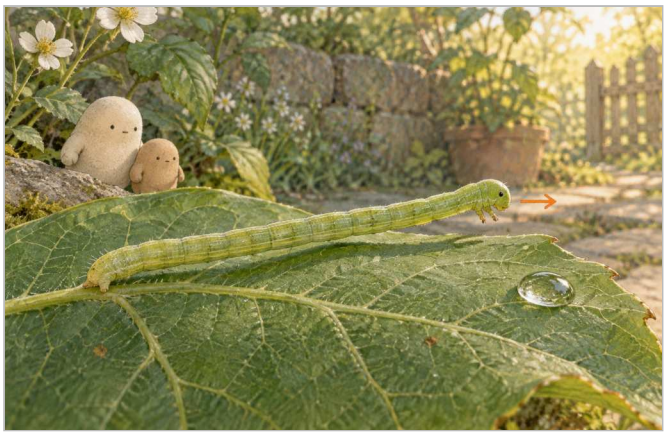
Front grips; rear comes closer. Notice the long legless middle rather than counting the partly hidden feet.

Look closer. Find the head at the right. Follow the body back to the rear feet. Where is there empty space underneath?

A little explanation. The front legs hold the leaf while the rear comes closer. The long middle bends upward into a loop. Inchworms have jointed legs near the head and fleshy gripping parts called prolegs near the rear, with a long gap between. They do not walk on a row of legs along that middle.

Try this. Draw a long curved line with its two ends close together. Imagine the front end stays in one place as the rear comes closer. Which part must bend?

02 What holds on while the head reaches?



Rear grips; front reaches. This simplified still picture shows one moment in a repeating movement.

Look closer. The front is lifted. Follow the body to the far left. Which feet still touch the leaf?

A little explanation. Now the rear prolegs anchor the body: they hold it in place. The front reaches ahead as the middle straightens. When the front legs grip again, the rear can catch up. The two ends take turns holding on. That repeating sequence carries the caterpillar across a leaf.

Try this. Look at the rear grip. What might happen if it slipped while the front was still reaching? Use a detail from the picture to explain.

03 How could a leaf vein show it moved?



Two imagined observations of the same leaf. Compare the head marks against the same fork.

Look closer. Compare the two leaf sketches. Each dot records the head. Find the same Y-shaped vein fork in both. What changed?

A little explanation. A bent body alone does not tell us how far the animal traveled. The first head mark is before the fork; the later mark is beyond it. The fork is a reference point, a place to compare positions against. Those changed positions are evidence for movement. They do not tell us its speed without times.

Try this. Draw a Y and put a dot before it. Draw the same Y again with a dot beyond it. Explain how the two drawings support the idea that the head moved.

04 Will this caterpillar always crawl?



An imagined later life stage. The notebook skips time; this change does not happen during one evening.

Look closer. Find three life stages in the notebook. How does the last shape differ from the first?

A little explanation. An inchworm is a caterpillar, the young stage of a geometer moth. It grows, becomes a pupa, then changes into a moth. This big change in body form is metamorphosis. It takes time; this scene imagines a later stage. A moth has six jointed legs. The caterpillar's fleshy prolegs do not remain.

Try this. Follow the notebook arrows: caterpillar, pupa, moth. Name one body feature that changes. What extra observation would help you learn how long the change takes?